

UNIT IV

6 Hours

CATALYSIS

LP II : Characteristics and applications of catalyst

Characteristics of Catalysts: The following are the general characteristics of a catalysts

- (i) A catalyst remains unchanged in mass and chemical composition at the end of the reaction

The amount and the chemical composition of a catalyst remain unchanged in a catalytic reaction. However, the catalyst may undergo a physical change. For example, manganese dioxide, which is used as a catalyst in thermal decomposition of potassium chlorate becomes powder during the course of the reaction.

- (ii) A small quantity of the catalyst is generally enough

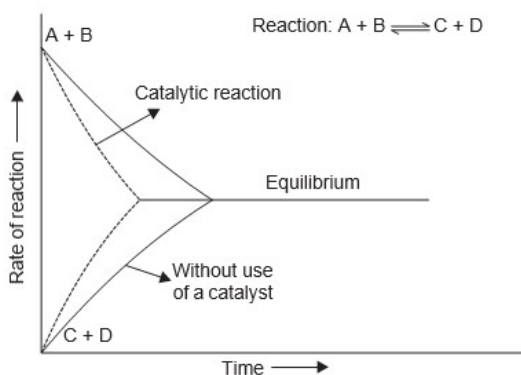
In most of the reactions, only a minute amount of catalyst is required. Only one gram of Cu^{2+} ions is sufficient to catalyse the oxidation of 10⁹ litres of sodium sulphite solution. In some cases, the rate of reaction is directly proportional to the concentration of catalyst present. Catalysis by acids or bases is usually of this type

- (iii) A catalyst is more effective when finely divided

In heterogeneous catalysis, the solid catalyst is more effective when in a state of fine subdivision than it is used in bulk. Thus a lump of platinum will have much less catalytic activity than colloidal or platinised asbestos. Finely divided nickel is a better catalyst than lumps of solid nickel.

- (iv) A catalyst does not alter the position of equilibrium state of a reversible reaction.

A catalyst allows the equilibrium to be reached faster. However, it does not alter the equilibrium composition of the reaction mixture. It is because, a catalyst increases the rates of forward and backward reaction equally.



Effect of catalyst on time to attain equilibrium.



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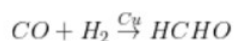
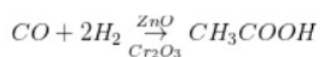
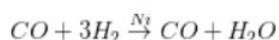
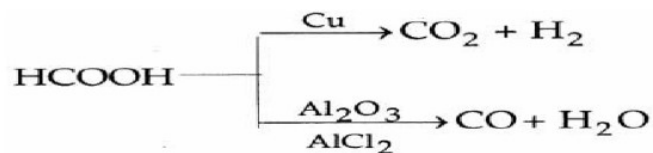
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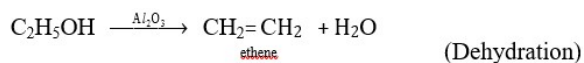
(v) Catalysts are generally specific in their action

The catalyst is specific in nature. It means by the change of catalyst, nature of the products changes or specific catalyst for a specific reaction.

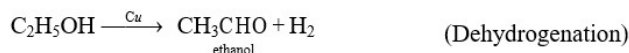


Generally, one catalyst will change the rate of only one reaction. For example, manganese dioxide catalyses the decomposition of potassium chlorate but not of potassium perchlorate.

Different catalysts, moreover, can bring about completely different reactions for the same substance. For example, ethanol ($\text{C}_2\text{H}_5\text{OH}$) gives ethene (C_2H_4) when passed over hot aluminium oxide,

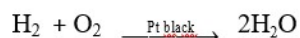
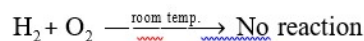


but with hot copper it gives ethanal (CH_3CHO)



(vi) A catalyst cannot initiate a reaction

A catalyst cannot, in general, initiate a reaction. In most cases a catalyst speeds up a reaction already in progress and does not initiate (or start) the reaction. But there are certain reactions where the reactants do not combine for very long period (perhaps years). For example, a mixture of hydrogen and oxygen, which remains unchanged almost indefinitely at room temperature, can be brought to reaction by the catalyst platinum black in a few seconds.





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- (vii) The activity of a catalyst can be increased by the presence of promoters and decreased by the presence of poisons.
- (viii) The catalyst exhibits maximum activity at a particular temperature which is known as optimum temperature. Change of temperature alters the rate of a catalytic reaction as it would do for the same reaction without a catalyst.

Important Applications of Catalysts

Reactions	Examples	Catalyst used
(i) Dehydration	$C_2H_5OH \longrightarrow C_2H_4 + H_2O$	Alumina, Zirconia
(ii) Hydrolysis	Vegetable oil + $H_2O \longrightarrow$ Fatty acid + glycerol	Twitchel's reagent
(iii) Oxidation	$2SO_2 + O_2 \longrightarrow 2SO_3$ $C_6H_6 + O_2 \longrightarrow$ Maleic anhydride $C_{10}H_8 + O_2 \longrightarrow$ Phthalic anhydride	V_2O_5
(iv) Hydrogenation	Oil + $H_2 \longrightarrow$ Vanaspati $N_2 + 3H_2 \longrightarrow 2NH_3$ (Haber's process) Alkene + $H_2 \longrightarrow$ Alkane	Raney Ni Fe Ni, Pt, Pd
(v) Dehydrogenation	$C_6H_5C_2H_5 \longrightarrow C_6H_5CH=CH_2 + H_2$ (Styrene)	Cr_2O_3/MoO_3
(vi) Alkylation	Friedel-Craft's reaction $C_6H_6 + RX \longrightarrow C_6H_5R + HX$	Anhydrous $AlCl_3$, BF_3 , HF
(vii) Isomerisation	Normal paraffin $\longrightarrow$ isoparaffin	Anhydrous $AlCl_3$
(viii) Cracking	Long chain paraffin $\longrightarrow$ branched chain alkanes	Bauxite, MoO_3
(ix) Halogenation	Benzene + $Cl_2 \longrightarrow$ Chlorobenzene	Fe or anhydrous $FeCl_3$
(x) Polymerisation	Ethene $\longrightarrow$ Polythene Styrene $\longrightarrow$ Polystyrene Vinyl chloride $\longrightarrow$ PVC	Zeigler catalyst or Peroxides or Persalts

Benefits and Limitations of Catalysis

Each type of catalysis has its own strengths and limitations. Homogeneous catalysis is highly reactive and selective, but separation of products from the catalyst remains challenging due to the homogeneity of the mixture. Heterogeneous catalysts are less reactive and selective, but the products are relatively easy to separate from the catalyst. Biocatalysts can be highly selective but typically have shorter lifetimes than other types of catalysts.

Questions

- Draw a distribution curve showing how the proportion of molecules able to react increases when a catalyst lowers the activation energy.
- "A catalyst doesn't affect the final state of equilibrium, although it shortens the time required to establish the equilibrium." - Justify
- Why do we prefer finely divided catalysts?